

# Isobutanol, Bench Side

Eleven Avalos-lab strains, and which of them a first RNA-seq round carries

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## Abstract

Eleven strains built for a branched-chain amino acid biosensor are on hand, labeled by tube number and strain number and by nothing else. Eight of them are named in the paper that built them, and those eight turn out to be screening hosts and single-allele controls rather than the strains that produced the paper's headline titers: three of those headline strains are a single transformation past a tube already in the freezer. Three more tubes are named nowhere in the paper's running text at all. For sequencing, the panel spans roughly an order of magnitude in isobutanol titer, that span is not a clean ladder because the carbon source changes between arms, and none of these strains shares a genetic background with the knockout-collection isobutanol work they get grouped with. A five-strain first round follows from that, anchored on the screening hosts rather than on a wild type the panel does not contain.

**How to read this.** Section 1 is the mapping from tube to strain, and it is the part that has to be right before anything else is decided. Section 2 is the relationship to the dissertation work on the yeast knockout collection, which shares a biosensor cassette with these strains but not a genetic background. Section 3 is the first sequencing round that falls out of the two.

Source paper: Zhang *et al.*, *Nature Communications* **13**, 270 (2022), <https://doi.org/10.1038/s41467-021-27852-x>, mirrored as zhangBiosensorBranchedchainAmino2022.

Generating script: experiments/034-isobutanol-wetlab/scripts/strain\\_panel.py

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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## 1 What Is Actually in the Freezer ✗

All eleven tubes come from one paper: Zhang *et al.*, *Nature Communications* **13**, 270 (2022). It builds a transcriptional biosensor for branched-chain higher alcohol (**BCHA**) metabolism and then uses it to run three screens. The sensor is yeast-enhanced green fluorescent protein (**yEGFP**) under the *LEU1* promoter, which Leu3p activates when  $\alpha$ -isopropylmalate ( $\alpha$ -IPM) accumulates, and  $\alpha$ -IPM tracks flux through the branched-chain pathway.

**Two sensor configurations, and they are not interchangeable.** The paper uses the sensor in two forms, and which form a strain carries determines which product its fluorescence reports.

the constructs containing yEGFP-PEST/LEU4<sub>1–410</sub>/leu2 $\Delta$  (henceforth called the isobutanol configuration) and yEGFP/Leu4 $\Delta$ S547/LEU2/leu4 $\Delta$ /leu9 $\Delta$  (henceforth called the isopentanol configuration)

The isobutanol configuration uses a destabilized yEGFP (the PEST degren) and a truncated Leu4p, **LEU4**<sub>1–410</sub>, that has lost its leucine regulatory domain. The isopentanol configuration uses stable yEGFP and a full-length Leu4p carrying a single serine deletion at position 547, **Leu4 $\Delta$ S547**. Five of our tubes sit in the first, three in the second.

**The background is CEN.PK2-1C, not S288C.** This matters more than anything else in the table, and section 2 is about why.

**Table 1: The eleven strains on hand, and what each one is in the paper that built them.** Color groups are as the tubes are labeled. Titters are not comparable down the column: the carbon source differs by row and two rows are a different product. A dagger marks a value we back-computed from a reported fold change rather than one the paper prints.

| ID    | Tube                 | Group  | Config.     | What it is                                                                                                                                                                                                                                                                                                                                                | Titer (mg/L)     |
|-------|----------------------|--------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| C0043 | YZy311               | green  | isobutanol  | Strain A of Fig. 3c: the <i>ILV6</i> screening host carrying wild-type <i>ILV6</i> on a CEN plasmid, with no isobutanol pathway overexpressed.                                                                                                                                                                                                            | –                |
| C0044 | YZy313               | green  | isobutanol  | Strain C of Fig. 3c: wild-type <i>ILV6</i> on a CEN plasmid in YZy363, which carries the mitochondrial isobutanol pathway $\delta$ -integrated. The denominator of the 1.6-fold improvement.                                                                                                                                                              | –                |
| C0045 | YZy314               | green  | isobutanol  | Strain D of Fig. 3c: the ILV6V110E hit in the same pathway-overexpressing host, 1.6-fold over C0044. One point mutation separates the two.                                                                                                                                                                                                                | –                |
| C0046 | YZy148               | blue   | isopentanol | Screening host for the <i>LEU4</i> mutagenesis libraries. Carries the isopentanol configuration with the <i>LEU4</i> $\Delta$ S547 allele removed, so a plasmid-borne <i>LEU4</i> is the only source of the enzyme.                                                                                                                                       | –                |
| C0047 | YZy148+LEU4WT        | blue   | isopentanol | Wild-type <i>LEU4</i> control for the isopentanol screen: YZy148 retransformed with a plasmid carrying unmutagenized <i>LEU4</i> .                                                                                                                                                                                                                        | 201 <sup>†</sup> |
| C0048 | YZy148+LEU4deltaS547 | blue   | isopentanol | The published leucine-insensitive allele, a single Ser547 deletion, carried back into the screening host. This is the benchmark the evolved <i>LEU4</i> variants were scored against.                                                                                                                                                                     | 842 $\pm$ 23     |
| C0049 | YZy452               | orange | isobutanol  | Best cytosolic-pathway isolate from three rounds of FACS on $\delta$ -site integrants, and the host the <i>Ll_ilvD</i> plasmid library was then screened in.                                                                                                                                                                                              | 310 $\pm$ 15     |
| C0050 | YZy454               | orange | isobutanol  | Wild-type <i>Ll_ilvD</i> control: YZy452 carrying a CEN plasmid with the unmutated dehydratase. The denominator of the 3.1-fold improvement.                                                                                                                                                                                                              | 73 <sup>†</sup>  |
| C0051 | YZy469               | orange | isobutanol  | The <i>Ll_ilvD</i> hit, I433V, on a CEN plasmid. Same host and same carbon source as YZy454, so the pair is a clean one-allele contrast.                                                                                                                                                                                                                  | 227 $\pm$ 13     |
| C0052 | YZy91                | gray   | isobutanol  | Screening host for the <i>ILV6</i> mutagenesis libraries. <i>BAT1</i> and <i>BAT2</i> are deleted so valine and $\alpha$ -KIV cannot interconvert, and <i>ILV6</i> is deleted so the only Ilv6p present is the plasmid-borne one under test. The tube itself carries no ILV6; the screen's best hit, this host plus ILV6V110E, reached 378 $\pm$ 10 mg/L. | –                |
| C0053 | YZy502               | gray   | isobutanol  | Optogenetic host: triple <i>pdv</i> $\Delta$ with light-inducible <i>PDC1</i> under OptoEXP and the dark-inducible cytosolic pathway under OptoINVRT7, $\delta$ -integrated. One plasmid and two FACS rounds short of the paper's best isobutanol strain.                                                                                                 | –                |

*S. cerevisiae* strain CEN.PK2-1C (MATa *ura3-52 trp1-289 leu2-3,112 his3-1 MAL2-8c SUC2*) and its derivatives were used in this study

**The tubes are hosts and controls, not the headline strains.** Read down the titer column and the panel looks weaker than the paper does. That is not an accident of which tubes were shipped. Eight of the eleven are either the host a mutagenesis library was screened in or the wild-type-allele control that a hit was scored against. The strains the paper leads with are the hits, and three of them are one transformation away.

**Table 2: Three of the paper’s headline strains are one construction step past a tube already in the freezer.** Each row is a single transformation, and in two cases a FACS enrichment, away from a strain we hold.

| Strain                         | From         | Step                                                                                                                                                         | Titer (mg/L) |
|--------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| YZy312 (Strain B)              | C0052 YZy91  | transform with pYZ228, the CEN URA3 plasmid carrying PTDH3-ILV6V110E-TADH1; this completes the Fig. 3c 2x2 that C0043, C0044 and C0045 are three quarters of | –            |
| YZy505                         | C0053 YZy502 | transform with the 2 $\mu$ plasmid pYZ350 (two copies of Ec_ilmCP2D1-A1, one each of Ll_ilmDI433V, <i>ARO10</i> , Ll_adhARE1), then two rounds of FACS       | 681 $\pm$ 29 |
| YZy470                         | C0051 YZy469 | move Ll_ilmDI433V from the CEN plasmid to a 2 $\mu$ plasmid                                                                                                  | 443 $\pm$ 6  |
| YZy148 + <i>LEU4</i> mutant #6 | C0046 YZy148 | transform with the evolved <i>LEU4</i> plasmid (E86D K191N K374R A445T S481R N515I A568V S601A)                                                              | 963 $\pm$ 32 |

**The green group is three quarters of a 2 $\times$ 2.** YZy311, YZy313 and YZy314 (C0043–C0045) are named nowhere in the paper’s running text, which is why the tube labels give nothing away. Supplementary Table 1 places all three at once, and Fig. 3c is where they appear:

Isobutanol titers of YZy91 transformed with CEN plasmids containing ILV6<sup>WT</sup> (Strain A) or ILV6<sup>V110E</sup> (*ILV6* mutant #6, Strain B) compared to titers from a strain overexpressing the mitochondrial isobutanol pathway (YZy363) transformed with CEN plasmids containing ILV6<sup>WT</sup> (Strain C) or ILV6<sup>V110E</sup> (*ILV6* mutant #6, Strain D).

That is a 2 $\times$ 2: two hosts crossed with two *ILV6* alleles. C0043 is Strain A, C0044 is Strain C, and C0045 is Strain D. The fourth cell, Strain B, is YZy312, and it is not in the box.

**All eleven are yeast, and seven of them carry a plasmid.** Nothing in this panel is an *E. coli* strain. Every one of the eleven appears in Supplementary Table 1, which the source titles "Yeast strains used in this study", and the only *E. coli* in the paper is DH5 $\alpha$  as a cloning host, plus *Ec\_ilmC*, an *E. coli* gene expressed in yeast.

The distinction that does cut through the panel is episomal. C0046, C0049, C0052 and C0053 are chromosomal only. The other seven each carry one plasmid, and all seven plasmids are the same backbone, AmpR CEN *URA3*, built off the empty parent pYZ125. Three things follow, and none of them is visible in a genotype string.

- **They have to be held on uracil dropout.** A CEN *URA3* plasmid under no selection segregates, and a plasmid-bearing tube streaked on rich medium quietly becomes its own parent. The paper’s own methods grow these in "SC (or SC-ura)" for exactly this reason.
- ***URA3* is spent in all seven.** That is the marker a guide library wants, so none of the seven can host one without curing the plasmid first.
- **A plasmid is a separate request from a strain.** AmpR means the plasmids live in *E. coli* in the source lab. Asking for pYZ228 is not the same transaction as asking for a yeast strain, and it is the cheaper of the two when the combination can be rebuilt here.

The last point simplifies two items on the request list. pYZ125 is the empty vector, so the missing vector control is C0049 transformed with pYZ125 rather than a request for YZy453; and YZy312 is C0052 transformed with pYZ228 rather than a request for a strain.

**Titers down that column do not compare.** Four measurement conditions appear in table 1: 15 percent galactose, 15 percent glucose, 2 percent glucose in the dark, and a 48 hour fermentation scored for a different product. What survives that is the set of contrasts in which exactly one thing changes, and there are three of them.

- **C0050 against C0051**, isobutanol, 15 percent glucose. *Ll\_ilmD* wild type against *Ll\_ilmD*<sup>I433V</sup> on the same CEN backbone in the same host. 3.1-fold, the largest single-allele effect in the panel.

**Table 3: Genotypes, as Supplementary Table 1 prints them, and which strains carry a plasmid.** All eleven are *S. cerevisiae*; none is an *E. coli* strain. Parent names the strain each was built from, so a chain can be walked back to CEN.PK2-1C (MATa *ura3-52 trp1-289 leu2-3,112 his3-1* MAL2-8c SUC2). Seven of the eleven carry an episomal plasmid and four do not. Every plasmid is the same backbone, AmpR CEN URA3, off the empty parent pYZ125, so each of those seven must be held on uracil dropout or it segregates the plasmid and reverts to its parent, and each has URA3 spent. The plasmids themselves are propagated in *E. coli* DH5 $\alpha$ , which makes a plasmid a separate request from a strain.

| ID    | Strain               | Parent     | Plasmid | Genotype                                                                                                                                                                                                              |
|-------|----------------------|------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C0043 | Yzy311               | Yzy91      | pYZ127  | Yzy91, CEN URA3 plasmid (PTDH3-ILV6-TADH1)                                                                                                                                                                            |
| C0044 | Yzy313               | Yzy363     | pYZ127  | Yzy363, CEN URA3 plasmid (PTDH3-ILV6-TADH1)                                                                                                                                                                           |
| C0045 | Yzy314               | Yzy363     | pYZ228  | Yzy363, CEN URA3 plasmid (PTDH3-ILV6V110E-TADH1)                                                                                                                                                                      |
| C0046 | Yzy148               | CEN.PK2-1C | none    | CEN.PK2-1C, <i>his3::HIS3-PLEU1-yEGFP-TADH1 bat1</i> $\Delta$ :: <i>hphMX leu4</i> $\Delta$ :: <i>lox71-kanMX-lox66 leu9</i> $\Delta$ :: <i>lox71-natMX-lox66 leu2::LEU2</i>                                          |
| C0047 | Yzy148+LEU4WT        | Yzy148     | pYZ149  | Yzy148, CEN URA3 plasmid (PTDH3-LEU4-TADH1)                                                                                                                                                                           |
| C0048 | Yzy148+LEU4deltaS547 | Yzy148     | pYZ154  | Yzy148, CEN URA3 plasmid (PTDH3- <i>LEU4</i> $\Delta$ S547-TADH1)                                                                                                                                                     |
| C0049 | Yzy452               | Yzy449     | none    | Yzy449, $\delta$ -integration-PTDH3-Ec_ <i>ilvCP2D1-A1-TADH1</i> -[PTEF1-Ec_ <i>ilvCP2D1-A1-TACT1</i> ]. Yzy449 is <i>ilv3</i> $\Delta$ <i>tma29</i> $\Delta$ with PGAL10-Ll_ <i>ilvD</i> and PTEF1-Bs_ <i>alsS</i> . |
| C0050 | Yzy454               | Yzy452     | pYZ126  | Yzy452, CEN URA3 plasmid (PTDH3-Ll_ <i>ilvD</i> -TADH1)                                                                                                                                                               |
| C0051 | Yzy469               | Yzy452     | pYZ353  | Yzy452, CEN URA3 plasmid (PTDH3-Ll_ <i>ilvDI433V</i> -TADH1)                                                                                                                                                          |
| C0052 | Yzy91                | CEN.PK2-1C | none    | CEN.PK2-1C, <i>his3::HIS3-PLEU1-yEGFP-PEST-TADH1-PTPI1-LEU41-410-TPGK1</i> , <i>bat1</i> $\Delta$ :: <i>hphMX bat2</i> $\Delta$ :: <i>lox71-kanMX-lox66 ilv6</i> $\Delta$ :: <i>lox71-natMX-lox66</i>                 |
| C0053 | Yzy502               | Yzy487     | none    | Yzy487, $\delta$ -integration-PC120-PDC1-TACT1-PTDH3-Ec_ <i>ilvCP2D1-A1-TCYC1-PTEF1-Ll_<i>ilvDI433V</i>-TTPS1-PGAL1-S-Bs_<i>alsS</i>-TACT1</i>                                                                        |

- **C0044 against C0045**, isobutanol. *ILV6* wild type against *ILV6*<sup>V110E</sup>, same host, 1.6-fold.
- **C0047 against C0048**, isopentanol. *LEU4* wild type against *LEU4*  $\Delta$ S547, same host, with C0046 as the allele-free floor beneath both.

C0043 does not pair with C0044, although both carry wild-type *ILV6*: their hosts differ by the *bat2*  $\Delta$  *ilv6*  $\Delta$  deletions and by the mitochondrial pathway at once, so the two differ in more than one thing.

**There is no wild type in the box.** Every one of the eleven carries the biosensor cassette, and most carry deletions in *BAT1*, *BAT2*, *ILV6*, *LEU4*, *LEU9* or the three *PDC* genes. An unengineered CEN.PK2-1C reference is not among them, and a differential expression analysis wants one. It is a cheap thing to request alongside the next shipment.

## 2 The Knockout-Collection Work Shares a Cassette, Not a Background ✗

These strains get grouped with the isobutanol screen of the yeast gene-knockout collection (**YKOC**) that ran in the same lab, and the grouping is half right. The screen is chapter 3 of José de Jesús Montaña López’s dissertation, *Systems Metabolic Engineering of Isobutanol Production in Saccharomyces cerevisiae* (Princeton, November 2024, adviser Jose L. Avalos), mirrored as `lopezSystemsMetabolicEngineering2024`. It names the paper above as its reference 158:

We used a previously designed branched-chain amino acid biosensor to rapidly quantify flux changes in the isobutanol biosynthetic pathway<sup>158</sup>

**The cassette is the same construct, to the promoter.** The dissertation’s Supplementary Table 5 gives the sensor strain as

YKOC, *his3::HIS3-PLEU1-yEGFP-PEST-TADH1-PTPI1-LEU41-410-TPGK1*

which is the paper’s isobutanol configuration term for term: destabilized yEGFP read out of P<sub>LEU1</sub>, amplified by the truncated LEU4<sub>1-410</sub> under P<sub>TPH1</sub>. The *leu2*  $\Delta$  the configuration calls for comes free, because every strain in the knockout collection is *leu2*  $\Delta$ 0. So a strain carrying the isobutanol configuration in our box and a strain from that screen are reporting the same quantity through the same reporter.

**The background is where they part.** The screen runs in BY4741, which the same table gives as **S288C** MATA *his3* $\Delta$ 1 *leu2* $\Delta$ 0 *met15* $\Delta$ 0 *ura3* $\Delta$ 0. The paper runs in CEN.PK2-1C. These are different laboratory lineages, and the paper’s own choice of CEN.PK2-1C for higher alcohol work is itself the argument that the choice of lineage is not neutral for this phenotype.

The consequence for sequencing is specific. A differential expression contrast between one of our tubes and a strain from the knockout screen would confound the perturbation with the lineage, and no normalization recovers that. The two sets support a comparison of *direction*, not of expression level.

**None of the eleven is a strain from that screen.** The dissertation names no strain with a YZy prefix anywhere in its text, and the paper names none of the dissertation’s yJM strains. The nearest relative to the screen in our box is any tube carrying the isobutanol configuration, and what it shares is the reporter.

**What to ask for, if the screen background is the point.** The strain the screen was anchored on is **yJM1**, BY4741 carrying that same cassette at *HIS3*, and the collection itself is the biosensor-containing YKOC. Both are Avalos-lab strains and neither is in the shipment. If the transcriptomics plan is meant to connect to the knockout screen rather than to the biosensor paper, yJM1 is the request to make, and it is a different request from the one that produced these eleven tubes.

### 3 The First Sequencing Round ✗

The ask is a low-flux anchor, the best producer, and steps in between. The panel supplies two independent versions of that, both for isobutanol, and one sentence in the Methods is what turns the first of them into three points rather than two.

**The promoter that makes a ladder.** The dehydratase in YZy452 is not constitutive.

Because the wild-type copy of *Ll\_ilvD* in YZy452 is expressed from a  $P_{GAL10}$  promoter, growing the library in glucose ensures this wild-type copy is repressed, and only the *Ll\_ilvD* variants from the CEN plasmid library (inserted downstream of a  $P_{TDH3}$  promoter) are expressed.

YZy452 also carries *ilv3* $\Delta$ , so the native dihydroxyacid dehydratase is gone. In glucose, C0049 therefore has no dehydratase at all, C0050 supplies the wild-type bacterial enzyme from a CEN plasmid, and C0051 supplies the improved allele from the same backbone. Three levels of one step, one host, one carbon source.

**Round one, five strains.** Two matched contrasts, on two different steps of the pathway.

1. **C0049 YZy452** – floor. The cytosolic pathway with its dehydratase step switched off by the carbon source.
2. **C0050 YZy454** – wild-type *Ll\_ilvD* restored. The unimproved-allele anchor.
3. **C0051 YZy469** – *Ll\_ilvD*<sup>I433V</sup>,  $227 \pm 13$  mg/L, 3.1-fold over C0050 on the same medium. One point mutation from the row above.
4. **C0044 YZy313** – wild-type *ILV6* in the mitochondrial-pathway host, Strain C of Fig. 3c.
5. **C0045 YZy314** – *ILV6*<sup>V110E</sup> in that same host, Strain D, 1.6-fold over C0044. A second one-mutation contrast, on the upstream acetolactate synthase regulator rather than the dehydratase.

Adding **C0053 YZy502** as a sixth is worth its slot for one reason: it is the direct parent of YZy505, the paper’s best isobutanol strain at  $681 \pm 29$  mg/L, so sequencing it now is the baseline that makes the YZy505 comparison possible later.

**Grow the first three on glucose, and measure titer alongside.** The paper measured C0049 on 15 percent galactose and C0050 and C0051 on 15 percent glucose. Glucose is the right pick for the round, because it is what makes the ladder a ladder, and the consequence is that C0049 will not reproduce its published  $310 \pm 15$  mg/L: that was a galactose number, and in glucose its dehydratase is repressed. Titer belongs in the same experiment as the sequencing rather than carried across from the paper.

**What is not in the round.** The isopentanol arm, C0046 through C0048, is the cleanest three-point series in the box and C0048 holds the highest titer of anything here at  $842 \pm 23$  mg/L. It is the leucine branch. This program is the valine branch, so those three are round two, and they move up only if a dose-response in  $\alpha$ -isopropylmalate flux is wanted for its own sake.

C0043 YZy311 sits out because its nearest neighbor in the panel, C0044, differs from it by both host and pathway, so a contrast between them would not isolate anything. C0052 YZy91 sits out because it is the plasmid-free host of C0043 rather than a producer.

**Three requests worth making with the next shipment.**

- **YZy121**, which Supplementary Table 1 gives as CEN.PK2-1C carrying the isobutanol-configured biosensor and nothing else. It is the reference every strain in this panel is missing: same lineage, same reporter, no deletions and no pathway.
- **YZy312**, Strain B, which completes the Fig. 3c 2×2 that C0043, C0044 and C0045 are three quarters of. It is one transformation of C0052 with pYZ228.
- **YZy453**, YZy452 carrying an empty CEN URA3 plasmid. C0049 carries no plasmid while C0050 and C0051 do, so the plasmid itself and its selection marker are currently confounded with the dehydratase.

None of the three connects this panel to the knockout screen; for that reason section 2 names yJM1 instead.

## References